TEAM UNIQUE · BUILD STAGE · AI FOR FOUNDATIONAL LEARNING

Class Map

Build and Win Guide

Product, AI, offline design, testing, demo, video, deck and submission

Prepared 17 Oct 2026 | Deadline: confirm 1 Nov vs 4 Nov (see section 0)

> **Is guide ka maqsad:** aapko batana ki Class Map ko **kya** banana hai, **kaise** banana hai, **kya galat ho sakta hai**, aur submission mein **kya dikhana hai**. Samjhane wale hisse simple Hinglish mein hain. Jo cheezein judges ya README mein jayengi (scripts, prompts, slides), woh English mein hain, taaki seedha copy kar sako.

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0 Facts: verified vs verify karna hai

Maine research kiya. Neeche likha hai kya confirm hua aur kya aapko khud check karna hai. Is table ko dhyan se padho, kyunki judges galat claim pakad lete hain.

Topic	Research mein kya mila	Status
Claude aur audio	Prerecorded audio Claude ke file-upload workflow mein supported nahi bataya gaya. Practical tareeka: pehle alag speech-to-text, phir text Claude ko. (Third-party sources.)	Official API docs mein ek baar check karo
Sarvam STT	Docs mein Saaras v3 recommended hai, Saarika v2.5 legacy hai. Hindi code hi-IN. REST API 30 second tak ki audio ke liye, lambi audio ke liye Batch API. Output modes: transcribe, translate, verbatim, translit, codemix. Vendor ka claim: saaf audio par 95-98% accuracy.	Bachchon ki awaaz par accuracy khud measure karo. Pricing aur limits docs se lo
WhatsApp Cloud API	Meta ka test number sirf 5 verified recipient numbers ko free messages bhej sakta hai. Business verification mein 1-5 working days lag sakte hain. Verification ke bina 2 numbers aur 250 business-initiated templates/24h ki limit (Clickatell doc). Oct 2025 se Cloud API hi ek model hai.	Meta docs mein latest limits check karo
ASER 2024	Grade 3 government schools: 23.4% bachche Grade 2 ka text padh sakte hain (2022 mein 16.3%). Subtraction: 33.7% (pehle 25.9%). Grade 5: lagbhag 44.8% Grade 2 text padh sakte hain. 52.1% government primary schools mein 60 se kam bachche. ASER 2024 ne 649,491 bachche, 17,997 gaon, 605 districts cover kiye.	Press/secondary sources se hai. Exact figure + base (govt ya sabhi schools) + page number ASER report se lo
DPDP	Act ka section 9: 18 saal se kam ke bachche ka data = verifiable parental consent. DPDP Rules 14 Nov 2025 ko notify hue, 3-stage commencement. Rule 10 consent ke tareeke batata hai. Fourth Schedule schools ko limited exemption deta hai, sirf educational activities aur safety ke liye.	Legal advice nahi hai. Pilot se pehle NGO partner ya legal expert se poochho
Judging criteria	Submission form mein: AI use + technical strength, low-connectivity real-world use, user inputs + testing, roadmap. Exact weights mujhe nahi mile.	Build Stage Guidelines PDF khud padho
Deadline	Form: 1 Nov 11:59 PM IST. Email: 4 Nov 8 PM IST.	Organisers se likhit mein confirm karo. Tab tak 1 Nov target
Claude models	Is waqt API strings: claude-haiku-5-5, claude-sonnet-5-5, claude-opus-5-5.	Docs mein check karo. Naam aur price badalte rehte hain

Ek correction: pehle playbook mein likha tha “7 mein se 10 Grade 3 bachche Grade 2 text nahi padh sakte.” Research ke hisaab se government schools mein yeh lagbhag **3 mein se 4** hai (100 minus 23.4 = 76.6%). Slide par likhne se pehle ASER report se exact figure aur page number lo.

1 Telegram ya WhatsApp bot kyu nahi?

Seedha jawab: bot **bana sakte ho**. Mere paas jo submission instructions hain unmein bot mana nahi hai. Mera suggestion hai ki **main product web app (PWA)** ho, kyunki submission ka format uske liye bana hai. Bot ek optional extra ho sakta hai.

Submission ke 4 niyam jo faisla karte hain

- **Hosted link** jo evaluators poore evaluation window mein khol sakein.
- **Test credentials har role ke liye** (teacher, headteacher, admin jaise).
- **Frozen deployed version**: jo submit kiya wahi evaluate hoga.
- **Low-connectivity behaviour** evaluators khud test karenge, aur har cheez ka offline fallback batana hai.

Teeno options ki seedhi tulna

Baat	Web app (PWA)	Telegram bot	WhatsApp bot
Evaluator kaise test karega	Link kholo, login karo	Bot ka naam dhundo, Telegram account chahiye	Test number sirf 5 verified numbers ko message kar sakta hai. Evaluators ko jodna mushkil
Har role ke credentials	Aasaan (login + role)	Chat command se role: awkward	Awkward
Offline	Asli offline design possible (cache + queue)	Bot ka logic server par hai, offline kuch nahi chalta	WhatsApp message queue karta hai (achha), par logic server par hi hai
India ke teachers	Link kholna ya install karna padega	Bahut kam teachers use karte hain	Sabse familiar
Setup mehnat	Medium	Kam (BotFather se minutes mein)	Zyada: Meta app, webhook, verification 1-5 din, template rules
Confirm/correct, groups, dashboard	Sabse behtar screen	Sirf buttons aur text	Sirf buttons aur lists
Headteacher/officer view	Natural	Phir bhi web page chahiye	Phir bhi web page chahiye

Meri recommendation

1	**Main product = PWA** Mobile-first web app. Teacher, headteacher aur block officer ke login. Offline queue. Yehi evaluators test karenge.
2	**Backend channel-agnostic rakho** Saari logic (STT, alignment, level, groups, plan) API ke peeche ho. Phir koi bhi channel us API ko call kar sakta hai.
3	**Telegram adapter = optional (1 din)** Agar time bache: python-telegram-bot se chhota adapter jo voice note lekar wahi API call kare. Demo video mein “same engine, doosra channel” dikhao.
4	**WhatsApp = roadmap** Deck mein asli steps likho: Meta app, business verification, templates, aur ki production mein teachers ke liye yahi channel hoga.

Agar team phir bhi WhatsApp hi chahti hai: test number se sirf 5 numbers kaam karenge, to evaluators ke liye web mirror aur video backup zaroor rakho. Warna judges kuch test nahi kar payenge.

2 Jeetne ki strategy

Submission form ke hisaab se evaluators yeh cheezein dekhenge. Har ek ke saamne likha hai ki aap kya dikhayenge.

Evaluator kya dekhega	Class Map ka jawab	Proof kahan dikhao
AI/LLM kahan lagta hai aur kya karta hai, layers mein	5 layers: input, processing, guardrails, evaluation, output (section 5)	Deck slide 5 + README AI services list
Galat AI output ka kya hota hai	Confidence score, teacher confirm, code se marking, activity bank, JSON check	Deck slide 6 + live demo (low-confidence sample)
Low connectivity aur sasti phone	Offline queue, manual mode, cached plan, chhoti audio, compressed photo	Video mein airplane mode demo + test log
User inputs aur testing	2-3 teacher interviews, validation numbers, “You said, we did” table	Deck slides 8-9
Roadmap aur existing systems se jod	Channel adapters, state FLN dashboards, content integration, languages	Deck slide 12
Code mein AI ka asli use	README mein har AI service aur key ki list, clean repo, no secrets	GitHub repo
Ease of use	Hindi UI, bade buttons, 3-tap flow, undo, status messages	Video scene 4
Challenge fit	Learning-Level Visibility: har bachche ka asli level dikhana + kal ka plan	Pitch ki pehli line

Wo 8 cheezein jo winners ko alag karti hain

- **Demo kabhi crash nahi hona chahiye.** Pehle 60 seconds mein evaluator ko result dikhna chahiye.

- **Sach likho.** “78% agreement, noisy audio mein kamzor” ek jhoothe 95% se zyada izzat dilata hai.
- **Asli users.** 2-3 teachers ke asli quotes, naam ki permission ke saath.
- **Har AI step samjha sako.** Kyu 90% threshold? Galat hua to kya? Kaun marking karta hai?
- **Offline asli ho.** Video mein internet band karke dikhao.
- **Cost ka asli number.** API usage log se, andaaza nahi.
- **Privacy ka jawab ready.** Audio delete, surname nahi, aggregate-only officer view.
- **Chhota scope, gehri quality.** Reading + grouping + plan + progress. Maths baad mein.

3 Final product definition

Product ek line mein: **teacher 30 second ka voice note record kare, aur app bataye ki bachcha kis level par hai, class ke groups kya hain, aur kal 20 minute mein kya karwana hai.**

Roles aur test accounts (evaluators ke liye)

Role	Kya dekh/kar sakta hai	Demo login (placeholder, apne banao)
Teacher	Apni class, bachche add, assess, confirm/correct, groups, daily plan, progress	teacher.demo / PIN
Headteacher	Apne school ki sabhi classes ka summary. Bachche ka naam nahi, sirf counts aur movement	head.demo / PIN
Block officer / Admin	Schools ka aggregate. Chhote groups hide. Activity bank aur settings (admin)	officer.demo / PIN

Features: Must / Should / Could / Won't

Priority	Feature
MUST	Login 3 roles. Class + bachche add (first name/roll). Reading assessment: record, upload, STT, alignment, level. Teacher confirm/correct. Grouping. Daily 20-minute plan. Progress view. Offline record + queue. Manual level mode. Sample inputs. Hindi UI.
SHOULD	Maths slate photo (vision + code marking + confirm). Cost log. Aggregate dashboards with small-group hiding. Re-assessment reminders.
COULD	Telegram adapter. English UI toggle. Print-friendly plan. Second language (Marathi/Bengali) passage bank.
WON'T (roadmap)	WhatsApp production bot. Parent messaging. Full curriculum. Written-work grading. On-device AI.

Ease-of-use features (video mein dikhane ke liye)

- Hindi labels aur bade buttons. Kam text, zyada icons. Ek screen par ek kaam.
- Record button par bada timer (20-30 s) aur “dobara record karo” ka option.
- Har result ke saath ek tap mein “Sahi hai / Badlo”. Undo available.
- Status saaf: “Internet nahi hai. 3 recordings wait kar rahi hain.”
- Aaj ka plan ek screen par, 1 minute mein padha ja sake. Materials ki list: sirf aam cheezein.
- Bachche ko kabhi label nahi: “shabdon par kaam kar raha hai”, “weak” nahi.

4 Architecture aur tech stack

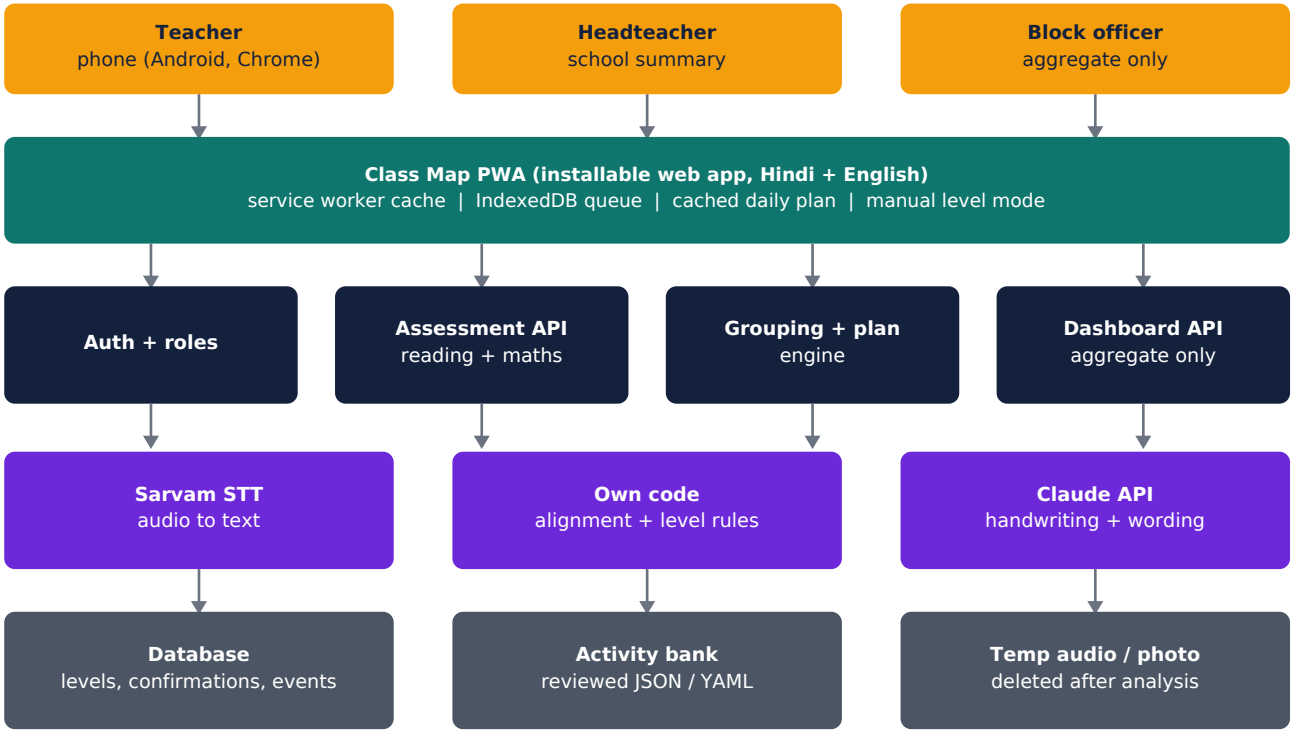


Figure 1: Class Map system architecture. Sab logic backend mein. Channel (PWA ya bot) sirf front hai.

Har hissa kya karta hai

Hissa	Kaam	Dhyan rakhne ki baat
PWA	Record, queue, confirm, plan, dashboards. Service worker se app offline khulta hai	Mic ke liye HTTPS zaroori. Android Chrome par test karo
Auth + roles	Login, role check (RBAC), school-wise data separation	Demo ke liye seeded accounts. Real data kabhi nahi
Assessment API	Audio/photo lena, AI services call karna, result save karna	Audio size limit. Audio store nahi, sirf level + evidence
Grouping + plan engine	Rules se groups aur activity bank se plan	LLM free-write nahi karta
Sarvam STT	Hindi audio se text	Bachchon par accuracy kam ho sakti hai. Fallback zaroori
Claude API	Slate handwriting padhna, error pattern aur plan ki simple wording	Sirf text aur images. Audio nahi
Database	Levels, AI vs teacher level, events, groups, plans	Teacher ka confirmed level alag column mein

Stack ka faisla

Layer	Recommendation	Kyu
Backend	Python + FastAPI	AI libraries aasaan. Tez likhne wala. Docs auto-generate
Frontend	Vite + (React ya plain JS) PWA, vite-plugin-pwa (docs check karo)	Service worker aur IndexedDB support. Jo aata hai wahi lo
Database	Local: SQLite. Hosted: Postgres (managed) ya SQLite persistent disk par	Chhota data. Simple rakho
Hosting	Always-on instance. Free tier jo so jaye (cold start) evaluation ke liye risky hai	Sasta paid tier ya Cloud Run min-instance 1. Pricing check karo
Audio capture	Browser MediaRecorder, mono, kam bitrate	Android Chrome aur iPhone Safari ke formats alag. Sarvam codec auto-detect karta hai, phir bhi dono par test karo
Secrets	Host ke environment variables. Repo mein sirf .env.example	Key leak ka sabse bada risk

5 AI design (judges yahin sabse zyada poochhenge)

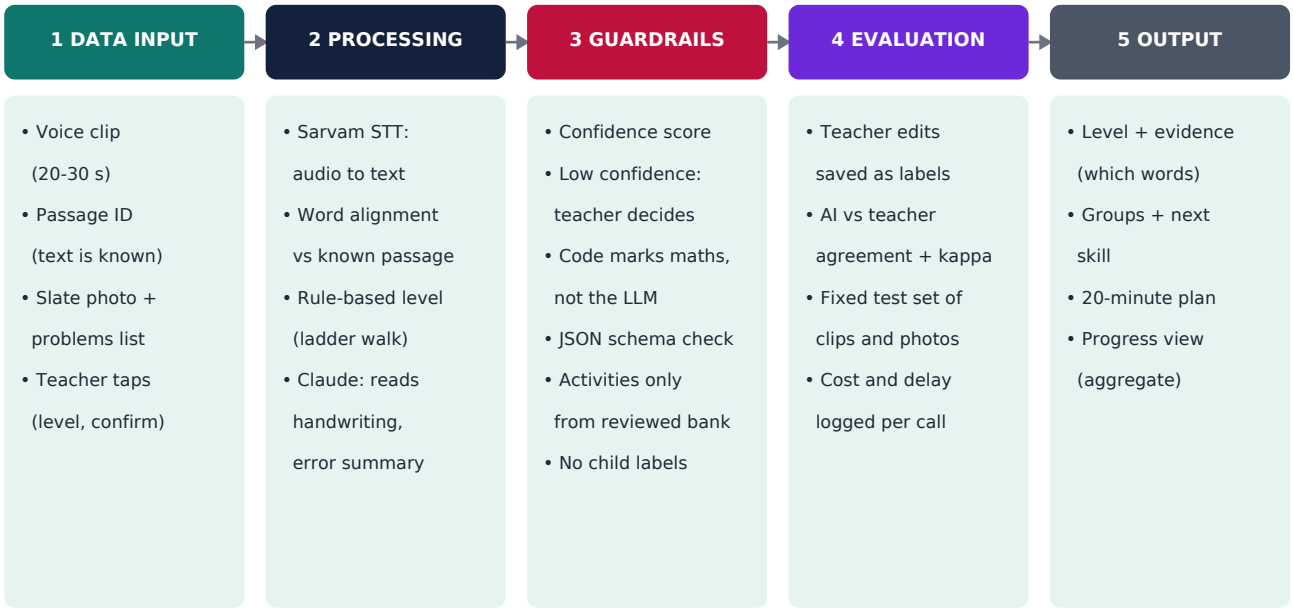


Figure 2: AI layers. Deck slide 5 mein yahi diagram lagao (judges ne yahi “layers” maange hain).

5.1 Speech-to-text (STT)

- **Claude audio nahi leta.** Isliye STT alag chahiye. Pehli choice: **Sarvam Saaras v3** (Hindi hi-IN). Mode **verbatim** aur **transcribe** dono try karo. Verbatim reading assessment ke liye zyada sahi lag sakta hai, par yeh aapko test se pata chalega.
- Comparison ke liye 1-2 aur options: Google Cloud STT (Chirp), Gemini audio input, AI4Bharat IndicConformer. Same 30 clips par chalao, sab ka agreement naapo, aur jo behtar ho wahi lo.
- Sarvam REST API **30 second** tak. Isliye har recording 30 s se chhoti rakho: passage ko 2 chhote clips mein todo ya Batch API use karo.
- Vendor ka 95-98% claim saaf audio ke liye hai. Bachchon ki awaaz aur classroom ke shor mein kam hoga. **Apna number measure karo aur wahi likho.**

5.2 Passage bank (aapka apna content)

- Har level ke liye 3-5 passages **khud likho**, original: letters (5-10), words (10), paragraph (Grade 1 level, 4-5 chhote vakya), story (Grade 2 level, 30 s se chhoti).
- Har passage ka ID, level, text, expected seconds, aur target sounds (jaise ch/chh) store karo. Taaki error pattern pakad sako.
- Kisi book ya TaRL material se copy mat karo. Ek teacher se padhwa ke check karwa lo ki bachche ke level ke hisaab se sahi hai.

id	level	text (original, Hindi)	sec	target sounds	reviewed by
P1	paragraph	...	20	ch, chh, ra	teacher name
S1	story	...	28	...	...

5.3 Alignment aur level decision (tested code Appendix A mein)

1	**Normalize** Devanagari text saaf: punctuation, danda hatao, nukta aur chandrabindu ek jaisa karo. Taaki spelling ke chhote fark galti na ginein.
2	**Align** Passage aur transcript ko word-by-word match karo (edit distance). Har word: correct, close, wrong, skipped, extra.
3	**Score** score = (correct + close) / total passage words. Strict score bhi store karo (sirf correct).
4	**Confidence** Transcript ki length, close words ka share, aur audio duration se 0-1 confidence. Kam ho to teacher faisla karti hai.
5	**Ladder walk** ASER jaisa: paragraph se shuru. Pass ho to story. Fail ho to word list. Phir letter. Isse har bachche ke 2 clips hi lagte hain, 4 nahi.
6	**Teacher confirm** AI level aur teacher level dono save. Difference = aapka accuracy number.

Thresholds placeholders hain (letter 0.8, word 0.8, paragraph 0.9, story 0.9). ASER tool documents aur teacher labels se confirm karo, phir deck mein apne final thresholds aur kaise nikale, yeh likho.

5.4 Claude kahan lagta hai (aur kahan nahi)

Kaam	Kaun karta hai	Model
Audio se text	Sarvam STT	Not Claude
Word match aur level	Aapka code	Not an LLM
Error pattern ka simple Hindi summary	Claude, sirf aligned data deke	claude-haiku-5-5 (sasta)
Slate ki handwriting padhna	Claude vision, JSON output	claude-sonnet-5-5
Maths marking	Aapka code	Not an LLM
Activity chunna	Rules + activity bank	Not an LLM
Activity ki simple wording	Claude, bank ke text se	claude-haiku-5-5

5.5 Maths slate photo

- Teacher ko photo guidance: acchi roshni, poori slate frame mein, ek bachcha ek photo, chehra frame se bahar.
- Photo bhejne se pehle phone par chhoti karo (jaise lambi side 1024 px, JPEG quality ~70, test karke dekho) taaki data kam lage.
- Claude sirf padhta hai (JSON). **Sahi/galat code tay karta hai.** Error type pehle rules se (smaller-from-larger, forgot-to-reduce-tens), phir plain words mein.
- Unreadable ho to **null**. Guess nahi. Teacher confirm.

5.6 Daily plan: activity bank se, free-write nahi

- 40-60 activities. Har ek ke tags: subject, level, skill, minutes, materials, group size, teacher-led ya independent, language.
- Rule: ek time par sirf ek group teacher ke saath. Baaki groups ki activity independent ho aur 3 se kam materials ho.
- Claude sirf wording simple karta hai. Output check karo: koi nayi material ya nayi activity na aaye (bank ke materials se match).

id	level	skill	min	mode	materials (draft, teacher review zaroori)
R-L1	letter	letter sounds	8	teacher-led	paper letter cards, bag
R-L2	letter	match sound to letter	8	pairs, independent	paper cards, pebbles
R-W1	word	build words	8	teacher-led	letter cards
R-W2	word	read word list	8	independent	words written on board
R-P1	paragraph	echo reading	8	teacher-led	5 sentences on board
R-S1	story	retell in pairs	8	pairs, independent	none
M-N1	1-digit numbers	number line jump	8	independent	chalk line, bottle caps
M-S1	2-digit	tens and ones bundles	8	teacher-led	sticks, rubber bands

id	level	skill	min	mode	materials (draft, teacher review zaroori)
M-B1	subtraction	borrowing with bundles	8	teacher-led	sticks, rubber bands

Yeh sirf starting examples hain, aam classroom ideas jo aap apne shabdon mein likho. Ek teacher ya NGO mentor se review karwao aur “reviewed_by” field bharo. Kisi proprietary material se copy mat karo.

5.7 Daily plan ka fixed template (20 minutes)

Time	Kya hota hai
0-2 min	Poori class ko ek simple instruction. Groups apni jagah baithte hain
2-10 min	Group A teacher ke saath (teacher-led). Group B, C independent activity
10-17 min	Group B teacher ke saath. Group A aur C independent
17-20 min	Group C ko 3 min check. Sabko ek line mein wrap-up

5.8 Guardrails ki poori list

- Teacher confirm/correct har level par. Teacher ka level final.
- Low confidence par AI result nahi, teacher se poochho.
- LLM ko raw audio nahi, sirf aligned data do. Taaki woh galtiyan invent na kare.
- LLM output JSON schema se validate karo. Galat ho to dobara try, phir fallback text.
- Maths marking code se. Activities bank se. Level thresholds rules se.
- Bachche ko label nahi. Koi open-ended chatbot nahi.
- Rate limit aur spending cap taaki demo par kharcha na bade.

5.9 Evaluation layer (jo deck mein dikhana hai)

- **Reading:** 30-60 clips. AI level vs teacher level. Agreement % aur Cohen's kappa, per-level confusion matrix.
- **Maths:** 40+ slates. Extracted answer vs asli answer.
- **Slices:** shor wale vs saaf clips, ladke vs ladkiyan, alag accents. Fark report karo.
- **Dev/test split:** thresholds half clips par tune karo, baaki half par report karo. Isse overfitting ka sawal band.
- **Ops:** har call ka latency aur token cost log karo.

Reading alignment, confidence (align.py | TESTED in sandbox)

```
"""Reading alignment: compare STT transcript with the known passage (Hindi/Devanagari)."""
import re, unicodedata
from dataclasses import dataclass

def normalize(text: str) -> list[str]:
    t = unicodedata.normalize("NFC", text)
    t = t.replace("\u0964", " ").replace("\u0965", " ")       # danda, double danda
    t = re.sub(r"[^\w\s\u0900-\u097F]", " ", t)                # drop punctuation
    t = t.replace("\u093c", "")                                # remove nukta (nukta variants)
    t = t.replace("\u0901", "\u0902")                          # chandrabindu -> anusvara
    return t.split()

def lev(a: str, b: str) -> int:
    prev = list(range(len(b) + 1))
    for i, ca in enumerate(a, 1):
        cur = [i]
        for j, cb in enumerate(b, 1):
            cur.append(min(prev[j] + 1, cur[j-1] + 1, prev[j-1] + (ca != cb)))
        prev = cur
    return prev[-1]

def sim(a: str, b: str) -> float:
    return 1 - lev(a, b) / max(len(a), len(b), 1)

@dataclass
class Op:
    kind: str            # correct | close | wrong | skipped | extra
    ref: str | None
    hyp: str | None

def align(ref: list[str], hyp: list[str], close_thr: float = 0.7) -> list[Op]:
    n, m = len(ref), len(hyp)
    INF = 10**9
    cost = [[INF]*(m+1) for _ in range(n+1)]
    back = [[None]*(m+1) for _ in range(n+1)]
    cost[0][0] = 0
    for i in range(n+1):
        for j in range(m+1):
            if i < n and cost[i][j] + 1 < cost[i+1][j]:           # skipped
                cost[i+1][j] = cost[i][j] + 1; back[i+1][j] = (i, j, "skipped")
            if j < m and cost[i][j] + 1 < cost[i][j+1]:           # extra
                cost[i][j+1] = cost[i][j] + 1; back[i][j+1] = (i, j, "extra")
            if i < n and j < m:
                s = sim(ref[i], hyp[j])
                c = 0 if s == 1 else (0.3 if s >= close_thr else 1)
                if cost[i][j] + c < cost[i+1][j+1]:
                    cost[i+1][j+1] = cost[i][j] + c
                    kind = "correct" if s == 1 else ("close" if s >= close_thr else "wrong")
                    back[i+1][j+1] = (i, j, kind)
    ops, i, j = [], n, m
```

```
    while (i, j) != (0, 0):
        pi, pj, kind = back[i][j]
        ops.append(Op(kind, ref[pi] if kind != "extra" else None, hyp[pj] if kind not in ("skipped",) else None))
        i, j = pi, pj
    return ops[::-1]

def summarize(ops: list[Op], n_ref: int) -> dict:
    c = lambda k: sum(o.kind == k for o in ops)
    return {"n_ref": n_ref, "correct": c("correct"), "close": c("close"), "wrong": c("wrong"),
            "skipped": c("skipped"), "extra": c("extra"),
            "score": (c("correct") + c("close")) / max(n_ref, 1),
            "strict_score": c("correct") / max(n_ref, 1)}

def confidence(summary: dict, hyp_len: int, ref_len: int, audio_sec: float) -> float:
    """0..1. Low when STT output looks unreliable. Tune on your labelled recordings."""
    ratio = hyp_len / max(ref_len, 1)
    conf = 1.0
    if ratio < 0.4 or ratio > 1.8: conf -= 0.5     # transcript length very different from passage
    if summary["close"] / max(ref_len, 1) > 0.25:  conf -= 0.2  # many near-matches: ambiguous
    if audio_sec < 3 or audio_sec > 60:           conf -= 0.2  # too short / too long
    return max(conf, 0.0)
```

Ladder walk and level decision (level.py | TESTED in sandbox)

```
"""Adaptive ladder walk (ASER-style). Thresholds are PLACEHOLDERS: confirm with ASER tool + teacher labels."""
LADDER = ["beginner", "letter", "word", "paragraph", "story"]
THRESH = {"letter": 0.8, "word": 0.8, "paragraph": 0.9, "story": 0.9}   # VERIFY / TUNE

def next_task(history: dict[str, float]) -> str | None:
    """history: task -> score already measured. Returns next task to record, or None if level decided."""
    if "paragraph" not in history: return "paragraph"
    if history["paragraph"] >= THRESH["paragraph"]:
        return "story" if "story" not in history else None
    if "word" not in history: return "word"
    if history["word"] >= THRESH["word"]: return None
    return "letter" if "letter" not in history else None

def decide(history: dict[str, float]) -> str:
    for task in ("story", "paragraph", "word", "letter"):
        if task in history and history[task] >= THRESH[task]:
            return task
    return "beginner"
```

6 Offline aur low-data design

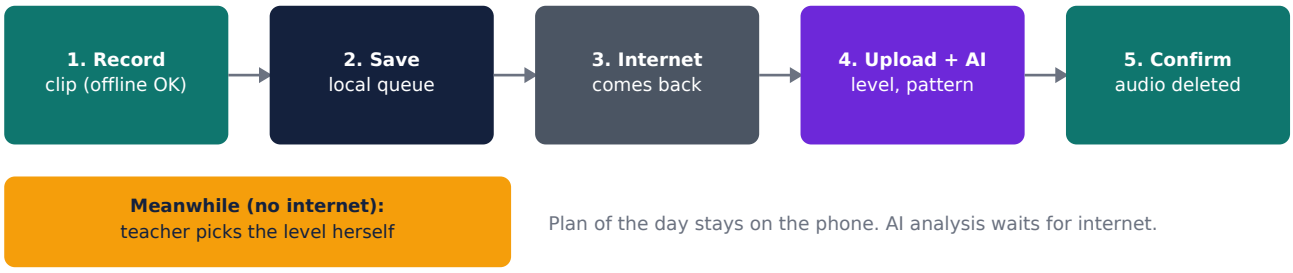


Figure 3: Offline flow. Recording aur manual level offline chalte hain. AI analysis internet aane par.

Kya offline chalega, kya nahi (deck mein yahi table dikhao)

Feature	Internet chahiye?	Offline mein kya hota hai
App khulna	Pehli baar haan	Baad mein cache se khulta hai
Bachche add / class dekhna	Nahi	Local copy se chalta hai, baad mein sync
Voice recording	Nahi	Phone mein queue mein save
AI level + error pattern	Haan	“Internet aane par process hoga” message
Manual level chunna (teacher)	Nahi	Poori tarah offline
Aaj ka plan padhna	Nahi	Pichla plan cached
Naya plan / groups	Haan (ya local rules se)	Groups local code se bhi ban sakte hain (roadmap)
Dashboards	Haan	Last synced data dikhao + “purana data” label

Low-data tips

- Audio mono, kam bitrate. 30 s ka clip kitne KB ka hai, yeh khud naapo aur deck mein likho (maine andaaza nahi diya).
- Photo upload se pehle resize + compress. Aur network par ek hi baar bhejo (idempotency key) taaki duplicate na ho.
- Retry with backoff. Timeout par clip queue mein rahe, delete nahi.
- iPhone Safari mein Background Sync nahi hota. Isliye app khulte hi aur “online” event par sync karo.
- Upload safal hone ke baad hi phone se audio delete karo.

Test kaise karein (evaluators bhi yahi karenge)

- Chrome DevTools: Network “Slow 3G” aur “Offline”. CPU throttling 4x-6x.
- Sabse sasti Android phone par asli test. Mic permission, audio format, Hindi font, battery.
- Lighthouse PWA audit chalao aur score/screenshot deck mein rakho.
- Ek “test log” banao: scenario, result, fix. Yeh “real scenario” ka proof hai.

Service worker (app shell cache) **(sw.js | UNTESTED skeleton: needs your keys/setup)**

```
// service-worker.js  (UNTESTED skeleton) : cache the app shell so it opens with no internet
const SHELL = "classmap-v1";
const FILES = ["/", "/index.html", "/app.js", "/styles.css", "/manifest.json"];
self.addEventListener("install", e => e.waitUntil(caches.open(SHELL).then(c => c.addAll(FILES))));
self.addEventListener("fetch", e => {
  if (e.request.method !== "GET") return;
  e.respondWith(caches.match(e.request).then(hit => hit || fetch(e.request)));
});
```

Offline queue with IndexedDB (queue.js | UNTESTED skeleton: needs your keys/setup)

```
// app.js : offline queue (UNTESTED skeleton). iOS has no Background Sync, so also sync when the app opens.
const db = await new Promise((ok, no) => {
  const r = indexedDB.open("classmap", 1);
  r.onupgradeneeded = () => r.result.createObjectStore("queue", { keyPath: "id", autoIncrement: true });
  r.onsuccess = () => ok(r.result); r.onerror = no;
});
const tx = (mode) => db.transaction("queue", mode).objectStore("queue");

export function saveClip(childId, task, blob) {              // works offline
  tx("readwrite").add({ childId, task, blob, status: "pending", ts: Date.now() });
}
export async function syncQueue() {                           // call on 'online' event + app start
  if (!navigator.onLine) return;
  const items = await new Promise(ok => { const q = tx("readonly").getAll(); q.onsuccess = () => ok(q.result); });
  for (const it of items.filter(i => i.status === "pending")) {
    const fd = new FormData(); fd.append("audio", it.blob); fd.append("child_id", it.childId); fd.append("task", it.task);
    const res = await fetch("/api/assess/reading", { method: "POST", body: fd, credentials: "include" });
    if (res.ok) tx("readwrite").delete(it.id);                // delete audio from phone after success
  }
}
window.addEventListener("online", syncQueue);
```

7 Privacy aur DPDP

Kya store hota hai	Kya store nahi hota
Bachche ka first name ya roll number. Level, evidence words, confidence, teacher level. Hash ki hui teacher phone.	Surname, DOB, ID number, chehre ki photo. Audio aur slate photo analysis ke baad delete.

- **Consent:** school aur teacher ke through. Asli bachchon ka data tabhi jab consent mile. Asli data repo mein kabhi nahi.
- **Third-party processing:** audio Sarvam ko aur photo Claude ko jaati hai. Dono ki data retention aur training policy khud padho aur README/privacy note mein likho.
- **Aggregate-only:** officer ko bachche ke naam nahi. 5 se kam bachchon ke groups hide (apni threshold chuno aur likho).
- **Bias:** accents, gender aur shor ke hisaab se accuracy alag naapo aur report karo.
- **DPDP:** Act ka section 9 (bachchon ka data) aur Rule 10 (verifiable consent) padho. Schools ke liye Fourth Schedule mein limited exemption hai, sirf educational kaam aur safety ke liye. Class Map kaun “data fiduciary” hai (school, NGO ya aap), yeh sawal pilot se pehle kisi legal expert se tay karwao. Yeh legal advice nahi hai.

8 Claude credits aur cost model

- **$500 credits** ka use: (1) coding assistance, (2) slate photo reading, (3) error summaries, (4) activity wording, (5) testing. Open-source consent par +250 credits.
- Sasta model pehle: text kaam ke liye **haiku**, vision ke liye **sonnet**. Opus ki zaroorat nahi lagti.
- API key sirf server par. Console mein agar spend limit set ho sakta hai to set karo. App mein rate limit lagao.
- Har response ka **usage** (input/output tokens) aur latency log karo. Yehi aapka asli cost data hai.
- Evaluator demo ke liye sample results cache karo (“cached” label), aur “live run” par limit lagao.

Cost per child per year: bharne wali table

Item	Kitna	Price (docs se bharo)	Cost
Reading clips per baseline (ladder walk avg)	~2 (naapo)	STT price/minute	...
Claude text tokens per clip	naapo (usage field)	Haiku input/output price	...
Maths photo per baseline	~1	Sonnet image + tokens price	...
Baseline + re-assessments per year	1 + 3 (assumption)		...
Hosting + database per month / bachche	school size se	Hosting price	...
Total per child per year			**Rs ...**

Formula: (baseline + re-assessments) x (clips x cost per clip + photos x cost per photo) + hosting share. Prices official pricing pages se lo, aur deck mein arithmetic dikhao.

9 Testing aur validation

Yeh section deck ke “User Inputs and Testing (1-2 pages)” ka saara saman hai.

1	**Reading study (30-60 clips)** Volunteers (consent ke saath): bachche agar mil jayein, warna bade log jo dheere ruk-ruk kar padhein. Har clip ko label karo (kaun, noise level). 1-2 teachers se level label karwao.
2	**Compute** Agreement %, Cohen's kappa, confusion matrix. Half clips par thresholds tune, baaki par report. Slices: noise, gender, accent.
3	**Maths study (40+ photos)** Slates ya notebook pages. Extracted vs asli answer. Unreadable ka share bhi report karo.
4	**Teacher interviews (2-3)** Mock-up ya demo dikhao. Quotes note karo (permission ke saath). Niche ka script use karo.
5	**Usability test** 3-5 log. Tasks: login, bachcha add, record, confirm, plan kholna. Time aur atakne ki jagah note karo.
6	**Low-connectivity log** Slow 3G, offline, sasti phone. Scenario, result, fix.
7	**“You said, we did” table** Har feedback ke saamne kya badla. Yeh deck ka sabse asli evidence hai.

Teacher interview script (10 sawal)

- Aap abhi bachchon ka level kaise pata karti hain? Kitna time lagta hai?
- Ek class mein kitne alag levels hote hain? Aap unke liye kya karti hain?
- Is app ko 1 bachche par aazmane mein kitna time lagega? Kitna theek lagega?
- Result dekh kar kya aap “Sahi hai / Badlo” karengi? Kab nahi karengi?
- Kal ka 20-minute plan: kya yeh asli class mein ho sakta hai? Kya nahi?

- Kaun si cheez samajh nahi aayi ya atki?
- Internet kaisa rehta hai school mein? Phone kaisa hai?
- Hindi mein kaun se shabd badalne chahiye?
- Kya aap isse har hafte use karengi? Kya rok sakta hai?
- Ek cheez jo hum zaroor badlein?

Recording ka consent line (apne shabdon mein badalo)

> “Hum Class Map naam ka ek hackathon project test kar rahe hain. Bachche ki awaaz sirf padhai ka level jaanchne ke liye record hogi, naam sirf pehla naam ya roll number rahega, aur analysis ke baad recording delete ho jayegi. Aap kabhi bhi mana kar sakte hain.” Consent school/teacher ke through, aur jahan zaroori ho abhibhavak ki sahmati ke saath.

Agreement and Cohen's kappa (metrics.py | TESTED in sandbox)

```
from collections import Counter
def agreement(a: list, b: list) -> float:
    return sum(x == y for x, y in zip(a, b)) / len(a)

def cohen_kappa(a: list, b: list) -> float:
    n = len(a); po = agreement(a, b)
    ca, cb = Counter(a), Counter(b)
    pe = sum(ca[k] * cb[k] for k in set(a) | set(b)) / (n * n)
    return 1.0 if pe == 1 else (po - pe) / (1 - pe)
```

10 Demo design (evaluators ke liye)

Evaluators ke paas kam time hoga. Pehle 60 seconds mein unhe result dikhna chahiye. App ke andar “Try a sample” tab banao:

ID	Kya dikhata hai	Label
S1	Saaf recording. Word level. Teacher confirm	sample (adult reader)
S2	Paragraph level, 2 galtiyan, error pattern ka summary	sample (adult reader)
S3	Shor wali recording. Low confidence. App teacher se poochti hai	sample (staged noise)
S4	Slate photo. Borrowing error pakadta hai	staged slate
S5	Unreadable photo. null. Manual entry	staged slate

- **Seeded demo class:** 24 synthetic bachche ke levels, groups, plan, aur progress ke saath. Saaf likha ho “synthetic data”.
- Har sample par 1 tap mein “cached result” aur ek chhota “run live” option (limit ke saath).
- Test credentials README aur submission form dono mein. Har role ka ek.
- Demo URL ko sab se alag naye phone par, Slow 3G par aur incognito mein test karo.
- Uptime monitor (free tools hote hain) lagao taaki pata chale link down hua ya nahi.
- Frozen version: ek alag production deployment, jisme sirf tagged release deploy ho.

11 Repo aur submission engineering

```
class-map/
  backend/            FastAPI app, align.py, level.py, grouping.py, maths.py, ai/ (sarvam.py, claude.py)
  frontend/           PWA (service worker, IndexedDB queue)
  data/               passages.json, activities.json (reviewed), seed_demo.json (synthetic only)
  eval/               labelled clip metadata (no audio of real children), metrics.py, results.md
  docs/               architecture.png, privacy.md, test_log.md
  .env.example        placeholders only
  README.md           setup, run locally, AI services list, test credentials
  LICENSE
```

README mein zaroor ho (English)

```
# Class Map
One-line pitch. Link to live demo. Link to demo video.
## Test credentials   (teacher / headteacher / officer)
## Quick start        (install, set env, run backend, run frontend)
## Environment        (.env.example explained)
## External AI services and models   <-- REQUIRED by submission
| Service  | Model            | Used for                          | Key variable        |
| Sarvam   | saaras:v3        | Hindi speech to text              | SARVAM_API_KEY      |
| Anthropic| claude-haiku-5-5 | error summary, plan wording       | ANTHROPIC_API_KEY   |
| Anthropic| claude-sonnet-5-5| handwriting reading (slate)       | ANTHROPIC_API_KEY   |
## Guardrails summary    ## Privacy note    ## Evaluation results    ## Known limits
## License
```

Submission se pehle

- **No secrets:** .env file ko .gitignore mein pehle din daalo. Secret scanner (gitleaks jaisa) pre-commit par chalao. Key leak ho jaye to turant rotate karo, file delete karna kaafi nahi.
- **No real data** repo mein. Eval folder mein sirf labels/metadata.
- **Evaluator access:** jab Tech4Dev evaluator IDs milen, usi din read access do. Ya repo public rakho.
- **Freeze:** deadline se pehle ek git tag (jaise v1.0-submission). Uske baad koi commit nahi. Dev branch alag.
- **Open-source:** haan karoge to licence chuno (jaise MIT ya Apache-2.0), aur dekho ki activity bank mein kisi ka copyrighted content na ho. Haan par +250 credits.
- **Submit hours pehle**, aur submission ka screenshot rakho.

12 3-minute video script

Public YouTube video. Pehle 3 minute hi dekhe jayenge, aur live link fail hone par yahi fallback hai.

Time	Scene (screen)	Voiceover (English)
0:00-0:20	Photo/illustration of a multi-grade class, then title	“Sunita teaches 38 children across five grades. Their reading levels are very different, but she has no quick way to measure them.” (+ one ASER stat with source)
0:20-0:40	Solution overview, one diagram	“Class Map turns a 30-second voice note into each child's level, class groups and tomorrow's 20-minute plan.”
0:40-1:25	Live: teacher login, record a clip, result with evidence words	“The app knows the passage, so it aligns what the child read to the text. The teacher sees the level and the words, and confirms or corrects.”

Time	Scene (screen)	Voiceover (English)
1:25-1:50	Groups screen, daily plan screen	“Children are grouped by real level. Each group gets an activity from a reviewed bank, using cheap materials.”
1:50-2:15	Airplane mode on, record, queue message, internet on, sync	“No internet in the classroom? Recording works offline and syncs later. The teacher can also pick a level manually.”
2:15-2:35	Low-confidence sample, teacher takes over	“When the AI is unsure, it says so and the teacher decides. Maths is marked by code, not by the model.”
2:35-3:00	Progress dashboard, validation numbers, roadmap	“We measured agreement with teachers at X percent, and we show where it is weak. Next: pilot schools and more languages.”

- Hindi subtitles ya English captions lagao. Audio saaf rakho (phone mic bhi chalega, shaant jagah).
- Screen recording ke baad ek baar phone par bhi record karo, taaki asli UI dikhe.

13 Deck outline (short PowerPoint)

#	Slide	Kya hona chahiye
1	Title + pitch	Class Map, ek line, team
2	Problem	Sunita + 3 cited ASER stats (page numbers)
3	Solution flow	8 steps ek diagram mein
4	Product screens	4-5 asli screenshots
5	**AI/LLM usage in layers**	Figure 2: input, processing, guardrails, evaluation, output
6	**Guardrails**	Confidence fallback, teacher confirm, code marking, schema, bank
7	**Low connectivity and low-cost devices**	Offline table + test log + phone model + Lighthouse
8	**User inputs and testing (1)**	Interviews, quotes, usability results
9	**User inputs and testing (2)**	Validation numbers + “You said, we did”
10	Cost and sustainability	Cost per child with arithmetic, who pays
11	Privacy and ethics	Delete audio, no surname, aggregate-only, DPDP note
12	**Future roadmap (1-2 pages)**	Languages, WhatsApp channel, pilot, integration with existing state FLN/school systems (verify feasibility before naming any)
13	Team and ask	Skills, what support you need

14 Judges ke sawal aur jawab

Q: Why not just send audio to a big model?	Claude does not take audio, and a general model can invent what a child said. We align a transcript to the known passage with rules, so every score has visible evidence.
Q: How accurate is it on children's Hindi?	We measured X% agreement with teacher labels on N clips and report weak slices (noise, accents). When confidence is low the teacher decides.
Q: What if speech recognition is wrong?	Confidence score, low-confidence fallback to teacher, and every result is confirmed or corrected by the teacher. Corrections are logged as labels.
Q: Why a web app, not WhatsApp?	The evaluation needs a hosted link, role logins and offline behaviour. The backend is channel-agnostic, so WhatsApp is the production roadmap.
Q: How do levels match ASER?	Our ladder follows the ASER tool structure; thresholds were checked against ASER guidance and tuned on teacher labels (we state the values).
Q: What about privacy of children?	First name or roll only, no photos of faces, audio and photos deleted after analysis, officers see aggregates only, and we checked DPDP Act section 9 with an expert.
Q: Who is responsible if a level is wrong?	The teacher confirms every level. The tool suggests; it does not decide.
Q: What works offline?	Opening the app, recording, manual level, and the daily plan. AI analysis needs internet and queues until it returns.
Q: What does it cost per child?	About Rs X per child per year, computed from logged API usage (show arithmetic).
Q: Will teachers actually use it?	Interviews: N of M teachers said they would. Assessment takes about T minutes per child (measured).
Q: How is the plan generated? Can the LLM hallucinate activities?	No. Activities come from a reviewed bank selected by rules. The LLM only rewords them, and we check that materials match the bank.
Q: Is the maths marking reliable?	The model only reads handwriting into JSON. Code marks it. Unreadable gives null and the teacher enters it.
Q: How will it scale?	Stateless API, aggregated dashboards, channel adapters, and per-clip cost is low and measured.

Q: Who pays and how is it sustainable?	State education department or NGO programmes first, CSR support; the open-source release helps adoption.
Q: How did you build it? Did AI write the code?	We used Claude as a coding assistant, as allowed by the programme, and the team understands and can explain every component.
Q: What are the limits?	Child speech recognition, noisy rooms, handwriting quality, and only Hindi so far. We list them in the README and the deck.

15 Din-ba-din plan (17 Oct se 1 Nov, 4 Nov buffer)

Team roles (3 log): **A** backend + AI, **B** frontend + offline, **C** data, validation, content, video, deck. 2 log hon to B aur C ke kuch kaam A ke saath baantne padenge.

Date	Kaam
Sat 17 Oct	Participation confirm, WhatsApp group, mentor slots book (ek banda). Guidelines PDF padho. Repo + .gitignore + .env.example. Anthropic aur Sarvam keys. Passage bank v1 likhna shuru.
Sun 18	30 recordings + labels. STT compare script (2 options). Activity bank v1 shuru.
Mon 19	Real clips par alignment + level chalao. Numbers dekho. STT choice final.
Tue 20	Backend skeleton, auth + roles, DB, seed. PWA shell + login.
Wed 21	Reading flow UI: record, upload, result, confirm. (Technical mentor)
Thu 22	Offline queue + manual mode + sync. Claude error pattern.
Fri 23	Grouping + activity bank (20+) + plan page.
Sat 24	Dashboards (headteacher, officer). Teacher test #1. (Education mentor)
Sun 25	Maths photo + code marking (agar time hai). Teacher test #2, #3.
Mon 26	Validation runs, thresholds tune. Cost logging. Sample inputs (S1-S5).
Tue 27	Hardening: rate limits, errors, Hindi text review, low-end phone test.
Wed 28	**Feature freeze.** README. Deploy frozen candidate.
Thu 29	Video record. Deck draft.
Fri 30	Deck final. Bahar ke 2 logon se demo link test karwao. Fixes.
Sat 31	Final checks, evaluator access, git tag, backup copies.
Sun 1 Nov	Submit subah. (Agar deadline 4 Nov confirm ho gayi hai, to 2-4 Nov sirf buffer: koi naya feature nahi.)

Cut order agar time kam pade: (1) Telegram adapter, (2) maths photo, (3) English toggle, (4) dashboards ki polish. **Kabhi mat kaato:** reading flow, teacher confirm, grouping + plan, offline queue, validation numbers.

16 Risks aur difficulties

Risk	Kab dikhega	Kya karein
STT bachchon par kamzor	Day 2-3 ke test mein	Alignment, confidence, manual fallback, sach likho, ek ek number
Mic kaam nahi karta	Phone par	HTTPS zaroori. Permission screen. Android aur iPhone dono test
Audio format mismatch	iPhone vs Android	Dono par record test. Server par format check ya convert
Hosting so jati hai	Evaluator ke pehle click par	Always-on instance. Cold-start test. Uptime monitor
Evaluation ke time Claude/Sarvam down	Live demo mein	Cached sample results + video fallback
Cost overrun	Evaluation window mein	Rate limit, spend limit, cached samples, sasta model
API key leak	Kisi bhi din	Env vars, scanner, rotate
Offline bug (queue duplicate)	Sync ke time	Idempotency key. Test with airplane mode
Hindi font ya layout toot jaye	Sasti phone par	System fonts, bada size, asli phone test
Teacher feedback nahi mila	Last week	Abhi se 2-3 teachers (ya NGO contact) fix karo
Scope creep	Roz	Cut order follow karo. Feature freeze Wed 28
Code samajh nahi aaya	Finals Q&A;	Har module ek team member explain kare. Dry-run karo
Evaluator ko samajh nahi aaya	Pehle minute	Sample tab, 1-tap demo, clear labels

17 Final checklist

Done	Item
[ ]	Hosted link, always on, tested cold, Slow 3G and offline
[ ]	Test credentials for teacher, headteacher, officer (form + README)
[ ]	3-5 sample inputs work in one tap
[ ]	Frozen version tagged; no commits after deadline
[ ]	GitHub: evaluator read access; no secrets; no real data; .env.example
[ ]	README: setup, local run, AI services and keys list
[ ]	Video: public on YouTube, under 3 minutes, shows offline

Done	Item
[ ]	Deck: AI layers, guardrails, low-connectivity, user testing (1-2 pages), roadmap (1-2 pages)
[ ]	Validation numbers real, with weak spots
[ ]	Cost per child with arithmetic
[ ]	Open-source choice made as a team (+250 credits)
[ ]	Challenge selected: Learning-Level Visibility
[ ]	Submitted hours before deadline; screenshot saved

A Appendix A: code (core files tested in sandbox)

Neeche ke pehle 5 files maine chalakar test kiye (sample Hindi passage, grouping, maths marking, kappa). Baaki (Claude, Sarvam, API, service worker, queue) skeletons hain: aapki keys aur setup chahiye. Dhyan: PDF se copy karte waqt indentation check kar lena.

Grouping algorithm (grouping.py | TESTED in sandbox)

```
def group_children(levels: dict[str, int], min_size: int = 3, max_groups: int = 4) -> list[dict]:
    """levels: child -> level score 0..4. Returns groups [{'level':[..], 'children':[..]}]."""
    by = {}
    for child, lv in levels.items():
        by.setdefault(lv, []).append(child)
    groups = [{"levels": [lv], "children": sorted(ch)} for lv, ch in sorted(by.items())]
    def center(g): return sum(g["levels"]) / len(g["levels"])
    # merge groups smaller than min_size into nearest neighbour
    changed = True
    while changed and len(groups) > 1:
        changed = False
        small = [g for g in groups if len(g["children"]) < min_size]
        if not small: break
        g = min(small, key=lambda x: len(x["children"]))
        others = [o for o in groups if o is not g]
        near = min(others, key=lambda o: abs(center(o) - center(g)))
        near["levels"] += g["levels"]; near["children"] += g["children"]
        groups.remove(g); changed = True
    # too many groups: merge the closest pair, preferring small ones
    while len(groups) > max_groups:
        groups.sort(key=center)
        i = min(range(len(groups)-1),
                key=lambda k: (abs(center(groups[k+1]) - center(groups[k])),
                               len(groups[k]["children"]) + len(groups[k+1]["children"])))
        a, b = groups[i], groups[i+1]
        a["levels"] += b["levels"]; a["children"] += b["children"]; groups.pop(i+1)
    groups.sort(key=center)
    return groups
```

Maths marking (code, not LLM) (maths.py | TESTED in sandbox)

```
"""Marking is done by CODE, not by the LLM. The LLM only reads handwriting into JSON."""
def mark(problem: tuple[int, str, int], child_answer: int | None) -> dict:
    a, op, b = problem
    truth = a - b if op == "-" else a + b if op == "+" else None
    if child_answer is None:
        return {"correct": None, "error": "unreadable"}
    if child_answer == truth:
        return {"correct": True, "error": None}
    err = "other"
    if op == "-" and a >= 10 and b >= 10:
        # column-wise smaller-from-larger bug, e.g. 52-27 -> 35
        col = abs(a % 10 - b % 10) + 10 * abs(a // 10 - b // 10)
        if child_answer == col: err = "smaller_from_larger"
        elif child_answer == truth + 10: err = "borrowed_but_forgot_to_reduce_tens"
        elif child_answer == a + b: err = "added_instead_of_subtracted"
    return {"correct": False, "error": err, "truth": truth}

def level_from_results(results: dict[str, bool]) -> int:
    """results: skill -> passed. Ladder: 0 beginner,1 one-digit,2 two-digit,3 subtraction,4 division."""
    lvl = 0
    for i, skill in enumerate(["num1", "num2", "subtraction", "division"], 1):
        if results.get(skill): lvl = i
        else: break
    return lvl
```

Sarvam STT call (sarvam_call.py | UNTESTED skeleton: needs your keys/setup)

```
# UNTESTED sample. Endpoint/fields as in Sarvam docs (REST: audio up to 30 s). Re-check docs before use.
import os, requests
def transcribe_hi(audio_bytes: bytes, filename="clip.webm", mode="verbatim") -> str:
    r = requests.post("https://api.sarvam.ai/speech-to-text",
        headers={"api-subscription-key": os.environ["SARVAM_API_KEY"]},
        files={"file": (filename, audio_bytes)},
        data={"model": "saaras:v3", "language_code": "hi-IN", "mode": mode}, timeout=60)
    r.raise_for_status()
    return r.json()["transcript"]
```

Claude calls: error pattern and slate reading (claude_calls.py | UNTESTED skeleton: needs your keys/setup)

```
# UNTESTED sample: needs your ANTHROPIC_API_KEY. Check model names in the current docs first.
import os, json, base64, anthropic
client = anthropic.Anthropic(api_key=os.environ["ANTHROPIC_API_KEY"])   # never hard-code the key
TEXT_MODEL   = os.getenv("CLAUDE_TEXT_MODEL",   "claude-haiku-5-5")    # cheap: error summary, plan wording
VISION_MODEL = os.getenv("CLAUDE_VISION_MODEL", "claude-sonnet-5-5")   # handwriting on slate

ERR_SYSTEM = ("You help a Grade 1-5 teacher in India. You get ALIGNED reading data (word by word). "
              "Describe the 1-2 most common error patterns in simple Hindi. Use only the data given. "
              "If there is no clear pattern, say so. Never label the child (no 'weak'). "
              'Return ONLY JSON: {"pattern_hi": str, "examples": [str], "next_skill": str}')

def error_pattern(ops: list[dict]) -> dict:
    r = client.messages.create(model=TEXT_MODEL, max_tokens=300, temperature=0,
        system=ERR_SYSTEM, messages=[{"role": "user", "content": json.dumps(ops, ensure_ascii=False)}])
    return json.loads(r.content[0].text)           # wrap in try/except + validate keys in real code

def read_slate(image_bytes: bytes, problems: list[str]) -> dict:
    b64 = base64.standard_b64encode(image_bytes).decode()
    prompt = ("Read a child's handwritten arithmetic on a slate. Problems given: " + ", ".join(problems) +
              '. For each return {"problem","child_answer" (string or null if unreadable),'
              '"legibility":"clear|partial|unreadable"}. Do not correct the child. '
              "Do not guess unreadable digits. Return only a JSON list.")
    r = client.messages.create(model=VISION_MODEL, max_tokens=500, temperature=0,
        messages=[{"role": "user", "content": [
            {"type": "image", "source": {"type": "base64", "media_type": "image/jpeg", "data": b64}},
            {"type": "text", "text": prompt}]}])
    return json.loads(r.content[0].text)
```

FastAPI endpoint skeleton (api.py | UNTESTED skeleton: needs your keys/setup)

```
# main.py (UNTESTED skeleton) : FastAPI. Run: uvicorn main:app --reload
from fastapi import FastAPI, UploadFile, Form, Depends, HTTPException
from align import normalize, align, summarize, confidence
from level import decide
app = FastAPI(title="Class Map")

@app.post("/api/assess/reading")
async def assess_reading(audio: UploadFile, child_id: int = Form(...), task: str = Form(...),
                         user=Depends(current_teacher)):          # current_teacher: your auth dependency
    data = await audio.read()
    if len(data) > 2_000_000: raise HTTPException(413, "audio too large")
    passage = PASSAGES[task]                                      # from passage bank (per task/level)
    text = transcribe_hi(data)                                    # Sarvam STT
    ref, hyp = normalize(passage["text"]), normalize(text)
    ops = align(ref, hyp); s = summarize(ops, len(ref))
    conf = confidence(s, len(hyp), len(ref), audio_sec=estimate_seconds(data))
    pattern = error_pattern([o.__dict__ for o in ops if o.kind != "correct"]) if conf >= 0.5 else None
    save_assessment(child_id, task, ai_score=s["score"], evidence=[o.__dict__ for o in ops],
                    confidence=conf, pattern=pattern)             # audio itself is NOT stored
    return {"score": s["score"], "confidence": conf, "needs_teacher": conf < 0.5, "pattern": pattern}
```

B Appendix B: prompts (English)

Error pattern (system prompt)

```
You help a Grade 1-5 teacher in India. You get ALIGNED reading data (word by word).
Describe the 1-2 most common error patterns in simple Hindi. Use only the data given.
If there is no clear pattern, say so. Never label the child (no "weak").
Return ONLY JSON: {"pattern_hi": str, "examples": [str], "next_skill": str}
```

Activity wording (system prompt)

```
Rewrite this classroom activity in simple Hindi for a teacher of a multi-grade class.
Use ONLY the materials listed. Do not add new activities, materials, or steps.
Keep it under 60 words. Return ONLY JSON: {"instruction_hi": str}
```

Slate reading (user prompt)

```
Read a child's handwritten arithmetic on a slate. Problems given: {list}.
For each return {"problem","child_answer" (string or null if unreadable),"legibility":"clear|partial|unreadable"}.
Do not correct the child. Do not guess unreadable digits. Return only a JSON list.
```

C Appendix C: sources aur kya verify karna hai

Topic	Source (research mein dekha)
Sarvam STT models, modes, 30 s REST limit	docs.sarvam.ai (speech-to-text overview, Saaras, FAQ)
Claude aur audio	Third-party articles (sonix.ai) and Claude Code feature-request threads. Confirm in official Claude API docs
WhatsApp Cloud API test number, 5 recipients	developers.facebook.com WhatsApp Cloud API: Get Started
WhatsApp business verification, 250 messages/day	help.clickatell.com (Get business verified); blueticks.co (Aug 2026 article)

Topic	Source (research mein dekha)
ASER 2024 figures	IANS report (morungexpress.com, socialnews.xyz); tnpscthervupettagam.com summary. Use the ASER 2024 report for page numbers
DPDP Rules 2025, Rule 10, Fourth Schedule	ssrana.in article; dpdpa.com (Rule 12 notes); raysolute.com (schools guide); naiknaik.com. Not legal advice

Yaad rakho: yeh guide aapko raasta deta hai, par jeet aapke **asli numbers, asli teachers ke quotes aur chalte hue demo** se aayegi. Pehle reading flow banao, pehle hafte mein hi STT ka asli test karo, aur har haftay teachers se feedback lo.